

ORIGINAL ARTICLE

Effect of Fufang Xueshuantong Capsule (复方血栓通胶囊) on A Rat Model of Retinal Vein Occlusion*

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ABSTRACT **Objective:** To establish a retinal vein occlusion (RVO) animal model and observe the therapeutic effect of a Chinese herbal composition (Fufang Xueshuantong Capsule, 复方血栓通胶囊, FXC) in ischemic retinal disease. **Methods:** Fifteen adult male Sprague-Dawley rats underwent laser photothrombosis to induce RVO on their right eyes and were subsequently randomized to receive FXC (the intervention group, $n=7$) or placebo treatment (the control group, $n=8$). Fundus fluorescein angiography was performed after 2, 4 and 8 weeks of treatment. Real-time reverse transcription-PCR was used to quantify the mRNA expression of vascular endothelial growth factor (VEGF) and stromal cell-derived factor-1 (SDF-1). The main outcomes were the mRNA copies of VEGF and SDF-1 and the counts of RVO signs. **Results:** Laser photothrombosis procedure induced typical lesions of RVO, including hemorrhage, leakage, retinal detachment, capillary non-perfusion, filling defect of retinal vessels, and lateral circulation/dilation of small vessels. The retinal lesions were associated with an increased expression of VEGF ($P<0.05$). No significant change of SDF-1 expression was noticed. Compared with the control group, the intervention group had numerically fewer RVO lesions at week 2 (1.71 ± 0.76 vs. 3.50 ± 1.51 , $t=-2.82$, $P<0.05$). The benefit of intervention remained at weeks 4 and 8. **Conclusions:** A rat model of laser photothrombosis-induced RVO was established and an increase in the VEGF expression was observed in the retinal lesion. The FXC had therapeutic benefit in improving retinal lesions in the rat model of RVO.

KEYWORDS Fufang Xueshuantong Capsule, retinal vein occlusion, animal model, stromal cell-derived factor-1, vascular endothelial growth factor

Retinal vein occlusion (RVO) diseases are one of the most prevalent causes of ischemic retinal diseases⁽¹⁻⁴⁾. Severe complications arising from retinal venous obstruction, such as macular edema and neovascular glaucoma, often result in vision loss at late stage^(3,5). The current treatments for RVO, however, are unsatisfactory^(4,6). For example, vision in 40% of treated eyes was worse than 0.5, and in 12% it was worse than 0.1 at 3 years⁽⁴⁾. Some Chinese herbal medications with the effects of both improving blood circulation and promoting angiogenesis⁽⁷⁾ may have the therapeutic usage for this indication. In the current investigation, an animal model of RVO was established. The expression of pro-neovascularization factors, stromal cell-derived factor-1 (SDF-1) and vascular endothelial growth factor (VEGF), in the ischemic retinas were examined. The therapeutic effect of a Chinese herbal composition, Fufang Xueshuantong Capsule (复方血栓通胶囊, FXC), was tested in this animal model.

METHODS

Animals

Adult male Sprague-Dawley rats weighing 200 to 250 g [SCXK (Hu) 2007-0002 and SYXK

(Hu) 2007-0002)] were obtained from Department of Laboratory Animal Science, Fudan University, and handled in accordance with the Association for Research in Vision and Ophthalmology (ARVO) Statement for the Use of Animals in Ophthalmic and Vision Research.

Drugs, Reagents, and Devices

FXC (ID080401) was provided by Zhongsheng Pharmaceutical Co., Ltd., Guangdong, China. The main ingredients of the compound medication are *Panax notoginseng*, *Salvia miltiorrhiza*, *Astragalus membranaceus*, and *Scrophularia ningpoensis*

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and the ratio of each ingredient was 5:1:1.5:1.5, respectively.

Reagents, including 0.5% proparacaine (0812K) and 0.5% tropicamide-phenylephrine ophthalmic solution (Mydrin-P1136), were purchased from Alcon, Puurs, Belgium and Santen, Osaka, Japan, respectively. Pentobarbital sodium and rose Bengal were purchased from Sigma Chemical, St. Louis, MO, USA. TRIzol reagent was purchased from Invitrogen, Carlsbad, CA, USA. The solution containing 20% fluorescein sodium was purchased from Mingxing Pharmaceutical Co., Ltd., Guangdong, China.

Laser equipment, 532 nm diode laser (Viridis), was purchased from Quantel Medical, Clermont-Ferrand, France. A small contact lens for laser photocoagulation was purchased from Ocular Instruments, Inc., Bellevue, WA, USA. Heidelberg Retina Angiograph (HRA2) was manufactured by Heidelberg Engineering GmbH, Germany. PCR thermocycler (model PTC-100™) and Mastercycler® ep realplex 4 were purchased from MJ Research, Watertown, MA, USA and Eppendorf AG, Hamburg, Germany, respectively.

RVO Model

Prior to examination or treatment, 15 rats were anesthetized by intraperitoneal injection (0.2 mL/100 g) of 1.5% pentobarbital sodium. The eyes were topically anesthetized with 0.5% proparacaine, and pupils were dilated with 0.5% tropicamide-phenylephrine. Laser photothrombotic RVO was performed in the right eye. The left eye served as a control. Rose Bengal, 20 mg/mL in sterile saline, was injected into the tail vein (20 mg/kg), and within 3 min, retinal veins were irradiated 1.5–2.0 optic disk diameters (DD) away from the optic nerve with a 532 nm diode laser. The rat fundus was viewed directly through a small contact lens. Laser parameter settings were 75 μ m in diameter, 0.2 to 0.3 s exposure time, and 0.1 W power. Approximately 4 to 6 branches of retinal veins at the lower quadrants or nearly half of the retinal veins were occluded to generate an inferior hemi-retinal occlusion.

Intervention

Fifteen rats that had received laser photothrombosis were randomly assigned to two groups by a random number table, with 7 in the

intervention group and 8 in the control group. The rats in the intervention group were treated each day after laser shootings with FXC 0.03 g/100 g, which was prepared by mixing the powder from FXC with purified water (volume ratio 1:1.5) and administered via stomach lavage. The rats in the control group were given with purified water of the same volume.

Fundus Fluorescein Angiography

To observe the evolution of venous occlusion and related retinal response, retinal photography and angiography were performed with a HRA2 at 2, 4, and 8 weeks of treatment. Rats were anesthetized with intraperitoneal injection of 20% fluorescein sodium (0.2 mL/100 g) before angiography was performed. The retinal lesions on angiography were categorized into six classes, including hemorrhage, leakage, retinal detachment, filling effect of retinal vessels, capillary non-perfusion, and lateral circulation/dilation of small vessels. The RVO signs in each affected eye were counted and added up for each group and then divided by the number of affected eyes to yield the mean count of RVO signs for each group.

Real-time Reverse Transcription-PCR

SDF-1 mRNA was detected by real-time reverse transcription-PCR (RT-PCR). Seven rats from the intervention group and one rat from the control group were examined 4 or 8 weeks after laser photothrombosis. The contralateral retinas served as the self-controls.

cDNA Synthesis and Amplification

Retina total RNA was extracted by using TRIzol reagent. cDNA was synthesized from total RNA by using PCR thermocycler. Quantitative PCR amplification was performed with gene-specific primers by using Mastercycler® ep realplex 4. Forty cycles of amplification included in each cycle 95 °C 30 s of denaturing, 56 °C 30 s (for β -actin and SDF-1), or 55 °C 30 s (for VEGF) of annealing and 72 °C 40 s of extension.

The primer sequences were as follows: SDF-1, forward 5'-TAAGCCAGTCAGCCTGAGC-3' and reverse 5'-GCACACTTGTCTGTTGTTGCT-3'; VEGF, forward 5'-CAGCGACAAGGCAGACTATT-3' and reverse 5'-GTTGGCACGATTTAAGAGGG-3'; and β -actin, forward 5'-AACCCTAAGGCCAACAGTGAAAAAG-3' and reverse 5'-TCATGAGGTAGTCTGTGAGGT-3'.

Quantification of Gene Expression

A relative standard curve method was used to calculate the amplification efficiency according to the manufacturer's instruction. Briefly, the synthesized cDNA was quantified by optical density and then diluted by 10-fold; thus, copies of cDNA in each sample corresponded to 1×10^{10} , 1×10^9 , 1×10^8 , 1×10^7 , 1×10^6 , 1×10^5 , 1×10^4 , and 1×10^3 , respectively (Figure 1). After quantitative PCR amplification was performed with gene-specific primers, a standard curve was drawn from seven points corresponding to seven different concentrations of cDNA mentioned before. Once the standard curve was made, target gene copies could be located at the curve (Figure 1). For each sample, the relative amount of gene expression was read from its respective standard curve and then normalized by the corresponding copies of reference gene (β -actin).

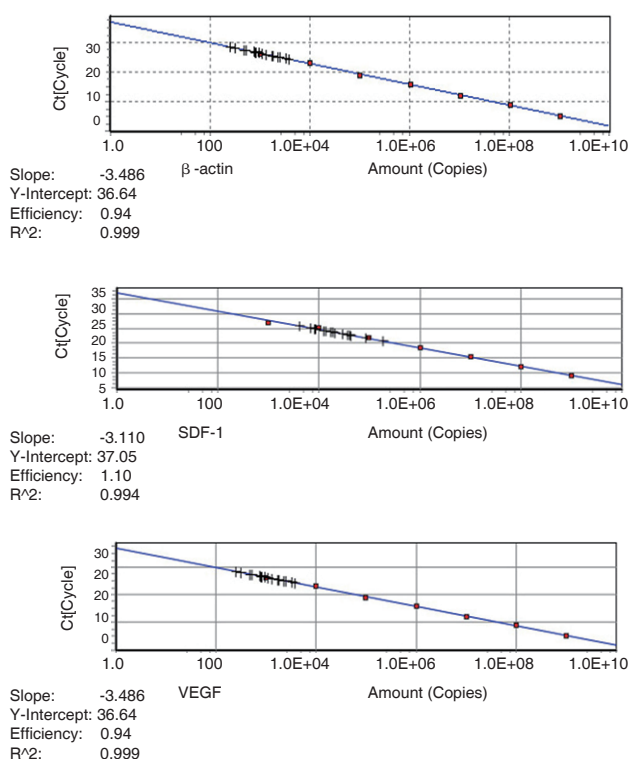


Figure 1. Standard Curves for β -actin, SDF-1, and VEGF in Real-time RT-PCR

Statistical Analysis

Student's *t*-test was used to compare the counts of RVO signs, and Wilcoxon signed-rank test was used to compare the copies of mRNA between groups. All analyses were performed by using SPSS (Windows version 10.0; SPSS, Inc., Chicago, IL, USA).

RESULTS

Manifestation of RVO

The vitreous and retinas were clear in the normal eyes on fluorescein angiographs. Typical lesions of RVO, such as hemorrhage, exudation, and edema, were shown on the retinas of the laser-irradiated eyes in the control group at 2 weeks after the laser irradiation. We also found the sign of delayed appearance of affected retinal arteries and/or veins. No obvious new vessels were found in all eyes (Figure 2).

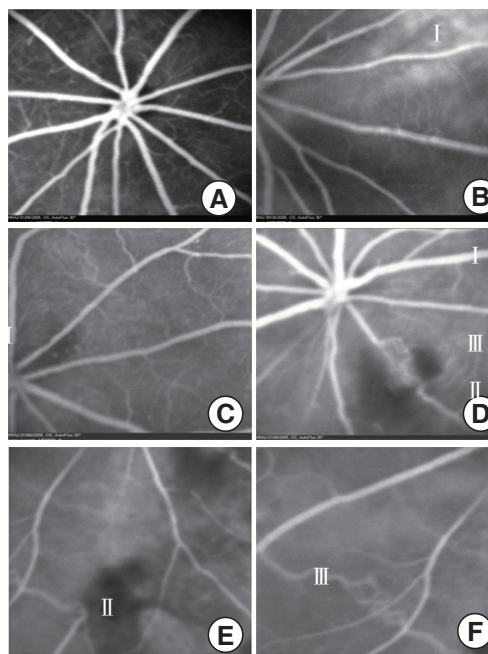


Figure 2. Retinal Fluorescein Angiogram Depicts the Lesions of RVO

Notes: A: angiograph of normal rat's retina; B to F: varied lesions of RVO; I: leakage/exudates; II: hemorrhage; III: lateral circulation/capillary dilation

Expression of VEGF and SDF-1 in RVO Retina

After normalizing to β -actin, the median copy number of VEGF cDNA was 5.16×10^{-2} in the RVO eyes and 4.20×10^{-2} in the contralateral, unmanipulated eyes (paired Wilcoxon signed-rank test, $Z = -2.10$, $P < 0.05$). No significant change of SDF-1 mRNA expression was detected by real-time RT-PCR in the RVO eyes.

Therapeutic Effect of FXC

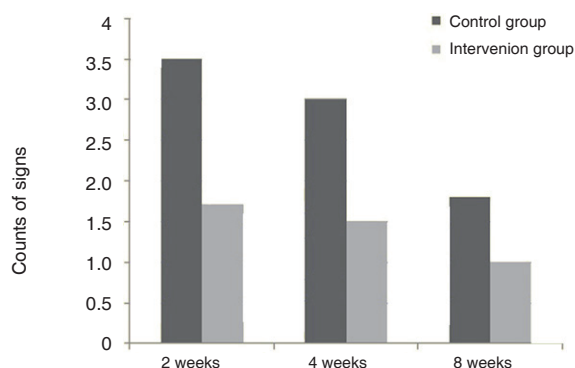
The RVO signs were quantified to assess the severity of the disease. Table 1 summarizes the RVO signs in the control and intervention groups. Four of the six signs, including hemorrhage, leakage,

Table 1. Counts and Percentages of the RVO Signs in Each Group at 2 Weeks after Laser Irradiation

Group	n	Count of signs ($\bar{x} \pm s$)	Percentages of each sign of RVO (%)					
			Hemorrhage	Leakage	Retinal detachment	Filling defect of retinal vessels	Capillary non-perfusion	Lateral circulation/ dilation of small vessels
Intervention	7	1.71 ± 0.76*	8.3	16.7	0	50.0	0	25.0
Control	8	3.50 ± 1.51	21.4	21.4	10.7	17.9	7.1	21.4

Note: * $P < 0.05$, compared with the control group

retinal detachment, and capillary non-perfusion, were scored lower in the intervention group. Overall, the intervention group had a lower mean count of RVO signs at 2 weeks compared with the control group (1.71 $\pm$ 0.76 vs. 3.50 $\pm$ 1.51, $t = -2.82$, $P < 0.05$, Table 1). As the retinal lesions improved over time (Figure 3), the counts of RVO signs were analyzed by the stratum of 4 weeks. The intervention group remained having fewer RVO signs (1.50 $\pm$ 0.71) compared with the control group (3.00 $\pm$ 1.00), although the difference was not statistically significant ($t = -1.80$, $P = 0.17$).

**Figure 3. Average Counts of RVO Signs**

Note: average counts of RVO signs in both groups were decreasing as time axis moving from 2 weeks after laser-induced RVO through 8 weeks

DISCUSSION

Retinal ischemia is a major cause of diabetic retinopathy, retinopathy of prematurity, and retinal arterial or venous occlusion⁽¹⁾. In the current investigation, we had successfully established an animal model of RVO that can be easily induced and reproduced. In clinical practice, RVO has a wide diversity of pathologic manifestations, including neovascularization, hemorrhage, exudation or leakage, filling defect of retinal vessels, capillary non-perfusion, dilation of small vessels, and lateral circulation⁽³⁾. In our rat model of RVO, all typical RVO signs, except for neovascularization, were shown on the affected retinas by angiography. As only half of

the retinal veins were occluded in the present model, the lack of severe ischemic environment in the model might partly explain why no new vessel formation was found herein. The degree of neovascularization among different RVO models varied by wide margins^(3,5,8-11), and the variation may be introduced by difference in the species' predisposition towards angiogenesis, numbers of vein branches occluded, and the sensitivity of the ways to detect new vessels in each study setting.

Other animal models for the study of ischemic retinal diseases have also been introduced. For example, diabetes models, either genetically modified or streptozotocin (STZ)-induced, are commonly used for the study of ischemia⁽¹²⁾. However, diabetes models are not only expensive and relatively inaccessible but also hard to maintain and with a high mortality rate. In addition, the underlying pathophysiologic changes of hyperglycemia and its comorbidities (e.g., hypertension and dyslipidaemia) could confound the ischemic condition. Therefore, diabetic animals could only be considered as a suboptimal model for the study of ischemic retinal diseases. The rat model in the current study, on the other hand, was easy to generate and reproduce and had nearly all the signs of ischemia without any comorbidity or complication. Thus, it can be an ideal tool to investigate non-proliferative ischemic retinal diseases.

Both SDF-1 and VEGF were considered as pivotal factors during ischemia and neovascularization^(12,13). VEGF is not only an important pro-angiogenesis/vasculogenesis factor but also a key factor that leads to the breakdown of blood-retinal barrier^(14,15). Thus, it was logical that the expression of VEGF mRNA was increased in the RVO retinas. SDF-1 is also considered as a crucial factor that promotes formation of new vessels in response to ischemia^(12,13). We have shown that the expression of SDF-1 was increased in the retinas of diabetic rats⁽¹²⁾. However, we did not

find a significant change of SDF-1 expression in the retinas of the RVO model in the present investigation. A possible explanation is that, as observed by Neuhaus⁽¹⁶⁾ and Butler⁽¹³⁾, SDF-1 is probably an upstream factor that regulates the expression of VEGF, and the increase in the expression of SDF-1 mRNA, if any, might occur earlier than that of VEGF and before the time point when we examined SDF-1 expression.

The treatment of RVO is still a subject of debate^(4,17-19). RVO in some patients are self-limiting^(5,20,21). In our model, when the rats were followed up from 2 to 8 weeks, the average count of RVO signs dropped from nearly 4 to less than 2. However, the risk of neovascularization as well as persistent hemorrhage, edema, or retinal detachment could lead to deterioration of visual acuity or sudden vision loss⁽³⁾. Current therapies include medication, laser retinal photocoagulation, intravitreal application of steroids or new VEGF inhibitors, and surgery. Available therapeutic efforts, as well as retinal laser photocoagulation, have shown limited success in improving visual acuity⁽¹⁷⁾. The unsatisfactory results of such therapeutic efforts led to the development of new treatment strategies⁽¹⁷⁾. The effectiveness of many new treatments, however, is unsupported by current evidence^(18,19). In our study, the compound herbal medication showed promising therapeutic effect as demonstrated by the significant reduction in RVO signs.

The main ingredients of the compound medication are *Panax notoginseng*, *Salvia miltiorrhiza*, *Astragalus membranaceus*, and *Scrophularia ningpoensis*—all are widely used in Chinese medicine (CM). An intervention study done on rats showed that *Panax notoginseng* was able to reduce bleeding time by half⁽²²⁾. The ability of *Salvia miltiorrhiza* to "thin" the blood by inhibiting platelet aggregation and reducing blood clotting is well documented⁽²³⁾. *Astragalus membranaceus* is utilized to tonify the qi and blood and is widely used as a tonic to enhance the body's natural defense mechanisms. It is also traditionally used to strengthen the immune system and in the healing of wounds and injuries⁽²⁴⁾. An *in vitro* experiment showed that the aqueous extracts of *Scrophularia ningpoensis* elicited significant inhibition of angiogenesis⁽²⁵⁾. Although data were encouraging in the current study, the compound

medicine's purported ability to invigorate the blood or improve circulation remained to be demonstrated in human trials.

There are two limitations to be highlighted in our study. First, as a preliminary study, we had a relatively small sample size that could weaken the power of statistical tests. The association between SDF-1 and RVO and the association between pro-neovascularization and the intervention might be underestimated. In addition, the small sample size also limited confounding adjustment, especially adjusting for the treatment duration. Second, we used the counts of RVO signs as the outcome measure to compare the treatment effect between the intervention group and the control group. Massive hemorrhage in vitreous or within retina could blur or make other lesions underneath invisible and a non-differential misclassification could potentially be introduced. Luckily, this was not the case in this study. Based on our findings, the prevalence of hemorrhage was much lower in the intervention group; the result would not likely be biased to exaggerate the treatment effect of the intervention.

In conclusion, we successfully established an easily induced animal model for ischemic retinal diseases and observed an increased expression VEGF in association with the retinal lesions. We further demonstrated the therapeutic effect of FXC for ischemic retinal diseases using the model. The findings of the current investigation add further to the notion that CM might be helpful to the ischemic retinal diseases; however, carefully designed studies are called for to validate these findings in human, and further studies are needed to clarify the therapeutic mechanism underneath.

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